Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/22/2021
PI Name:	Vimal, Vivekanand Ph.D.		
Project Title:	Predicting Individual Differences in Learning to Manually Stabilize Attitude in a Space Flight Analog Environment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02453-2728	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 TRI-RFA-17-01: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	12/01/2017	**End Date:**	11/30/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Period of performance changed at some point to 12/10/2017-11/30/2020 per TRISH submission in March 2020 (originally was 1/1/2018-12/31/2019) (Ed., 4/2/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	DiZio, Paul Ph.D. (MENTOR: Brandeis University)		
Grant/Contract No.:	NNX16AO69A-P0202		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP In Specific Aim 1, to characterize individual differences in subjects, we will use three tests of increasing complexity. The first test, in collaboration with Dr. Faisal Karmali measures each subject's vestibular thresholds. The second test measures each subject's spatial perception threshold by finding where a subject's ability to indicate the balance point deteriorates when exposed to different magnitudes of relevant gravitational cues. In the third test, in collaboration with Dr. Mark Shelhamer, we will examine performance in a manual tracking task which has been able to predict motor adaptation ability in a different paradigm. Next, we will determine whether any of these tests predict learning and performance in a space flight analog environment. In Aim 2, to study learning, generalization, and transfer of skill we develop two training programs to help subjects improve performance in our space flight analog. One training program examines incremental exposure to a condition without relevant gravitational cues and the other exploits our recent finding that balance control consists of two dissociable processes: identifying the direction of dynamic balance and identifying the direction of gravity. We study retention by asking subjects to come back a month later and perform the same dynamic stabilization task. These specific aims address NASA roadmap gaps: SM24 (Determine if the individual capacity to produce adaptive change (rate and extent) in sensorimotor function to transitions in gravitational environments can be predicted with preflight tests of sensorimotor adaptability) and Train-04 (We do not know the types of skills and knowledge that can be retained and generalized across tasks for a given mission to maximize crew performance).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have developed novel machine learning algorithms and techniques that can predict a person's performance in a disorienting space flight analog condition. These algorithms can be generalized to other situations. We have also developed an effective training program that enhances performance in 100% of people in the space flight analog condition. This training program could be integrated into astronaut training. This work lays the foundation to develop machine learning techniques that can help in real time training and space flight. In addition, this research can be used to develop technological countermeasures such as a more effective application of vibrotactile feedback.
Task Progress:	Space flights can cause many sensorimotor related difficulties that could jeopardize a mission. For example, if astronauts are forced to land manually onto the surface of Mars or the Moon, they will experience gravitational transitions while dynamically stabilizing the spacecraft. In low-g and 0g environments, gravitationally dependent vestibular and somatosensory cues are minimized and astronauts can easily become spatially disoriented. We strapped blindfolded subjects into a Multi-Axis Rotation System that was programmed to behave like an inverted pendulum. Subjects used a joystick to stabilize themselves around the balance point. When subjects balanced in the Vertical Roll Plane, where otolith and somatosensory gravitational cues about tilt were available, they performed well and showed robust learning across trials. By contrast, in the Horizontal Roll Plane, where relevant gravitational positional cues were absent and only semicircular canal cues about motion were available, they repeatedly crashed (exceeded angular displacements of 60) and showed minimal learning and persistent positional drifting. Because these subjects could not rely on gravitational cues to do the task, we call this our space flight analog condition. In this condition 90% of subjects reported spatial disorientation and 100% showed positional drifting in the data. In this project, we used machine learning to develop classifiers that could predict a subject's final performance in our space flight analog condition with 85% accuracy. We were also able to cluster individuals into Proficient, Somewhat-Proficient, and Not-Proficient groups and we discovered that the Not-Proficient subjects were not randomly bad, but instead, all of them had acquired the same suboptimal strategy when in the space flight analog condition. This suboptimal strategy consisted of using very large stereotyped joystick deflections which helped reduce the number of crashes at the cost of very large amplitude oscillations. As a countermeasure, we developed an effective training program that exploited our previous finding that balance control consists of two dissociable components (i.e., dynamic stabilization and alignment to the gravitational vertical), and it led to performance improvements in every subject. Even after 4 months, subjects retained the training with no performance decrements, suggesting that astronauts could be trained on Earth and rely on that training during space missions.
Bibliography Type:	Description: (Last Updated: 06/09/2025)
Articles in Peer-reviewed Journals	Vimal VP, Zheng H, Hong P, Fakharzadeh LN, Lackner JR, DiZio P. "Characterizing individual differences in a dynamic stabilization task using machine learning." Aerosp Med Hum Perform. 2020 Jun;91(6):479-88. https://doi.org/10.3357/AMHP.5552.2020 ; Pubmed PMID: 32408931 , Jun-2020
Articles in Peer-reviewed Journals	Vimal VP, DiZio P, Lackner JR. "The role of spatial acuity in a dynamic balancing task without gravitational cues." Experimental Brain Research. 2021 Oct 15;240:123-33. https://doi.org/10.1007/s00221-021-06239-w , Oct-2021
Articles in Peer-reviewed Journals	Wang Y, Tang J, Vimal VP, Lackner JR, DiZio P, Hong P. "Crash prediction using deep learning in a disorienting spaceflight analog balancing task." Front Physiol. 2022 Jan 28;13:806357. https://doi.org/10.3389/fphys.2022.806357 ; PMID: 35153834; PMCID: PMC8832067 , Jan-2022
Awards	Vimal VP. "1st place in NASA HRP conference poster competition, 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020." Jan-2020